

Predicting Air Turbulence Using Machine Learning

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Introduction

Air turbulence refers to the irregular and often unpredictable movement of air masses within the Earth's atmosphere. These turbulent air currents can cause aircrafts to experience sudden jolts, creating a bumpy ride for passengers.

Problem

Due to climate change, incidents of air turbulence have been on the rise in recent years, causing significant concern amongst passengers and airlines. Understanding air turbulence is crucial for passenger safety and comfort, flight efficiency, and operational planning. Being able to predict air turbulence would lead to safer flights and reduced operational costs.

Goal

The purpose of this analysis is therefore to examine the features influencing air turbulence, and to build a model that can predict air turbulence based on these features.

Audiences

Pilots, passengers, airlines, and air traffic controllers would benefit from this analysis:

- Pilots: Predicting turbulence would enable pilots to plan and adjust flight paths accordingly, thereby enhancing safety and efficiency
- Passengers: Predicting turbulence would reduce the discomfort and risk of injury faced by passengers, leading to a stress-free experience when flying
- Airlines: Knowing where turbulence might occur would help airlines plan flight routes to minimize delays and maintenance costs from wear and tear
- Air Traffic Control: Predicting turbulence would allow controllers to route flights in a safer and more efficient manner, leading to better management of airspace

Data

This analysis focuses on turbulence over U.S. mainland over the past 10 years (2015 – 2024).

The following data were sourced:

- Pilot reports (or PIREPs): Real-time turbulence reports filed by pilots
- Ground elevation: Considering ground elevation helps predict areas where turbulence is likely to occur; when strong winds flow over elevated terrain, the airflow is disturbed, creating turbulence
- Aircraft specifications: Aircraft type, weight class, engine type, and the number of engines can be critical factors in predicting turbulence because aerodynamic designs, structures, and capabilities affect how turbulence is experienced
- Meteorological conditions: Wind speed, air temperature, and humidity are critical in understanding the airflow around aircrafts that might lead to turbulence

The following table details the features and their respective sources:

Source	Feature	Column Name	Description	Data Type
Pilot Reports (PIREPs)	Date and time	DATETIME	Local date and time of report	Datetime
	Time of day	TIME_OF_DAY	When: Day, night, sunrise, or sunset	String
	Year	YEAR	Year of report	Integer
	Hour	HOUR	Hour of report	Integer
	Month	MONTH	Month of report	Integer
	Season	SEASON	When: Spring, summer, autumn, winter	String
	Flight level	FL	Flight level of incident (ft)	Integer
	Distance	DISTANCE_TO_GROUND	Distance to ground (ft)	Integer
	Aircraft	AC	Aircraft type designator	String
	Turbulence	TURB_CAT	Category of turbulence	String
	Severity	TURB_DEG	Severity of turbulence	Integer
	Longitude	LON	Longitude of report	Integer
	Latitude	LAT	Latitude of report	Integer
	State	STATE	State of location	String
	Urgent	URGENT	Whether the report was filed as urgent	Boolean
Elevation (USGS)	Elevation	ELEV	Ground elevation at location (ft)	Integer
Aircraft Specs (ICAO)	Manufacturer	MANUFACTURER	Manufacturer of plane	String
	Model	MODEL	Model of plane	String
	Description	DESCRIPTION	Type of plane	String
	Engine type	ENGINE_TYPE	Type of engine	String
	Engine count	ENGINE_COUNT	Number of engines	Integer
	Weight class	WTC	Weight class of plane	String
Meteorological data (NWS)	Wind speed	WIND	Daily average wind speed at location	Integer
	Wind direction	WIND_DRCT	Daily average wind direction at location	Integer
	Cardinal wind	WIND_DIR	Daily average cardinal wind direction	String
	Max temp	MAX_TEMP	Daily maximum temperature at location	Integer
	Min temp	MIN_TEMP	Daily minimum temperature at location	Integer
	Humidity	HUM	Daily relative humidity at location	Integer

**cells in yellow denote engineered features*

Patterns, Trends, and Insights

Exploratory Data Analyses (EDA) were performed in Python to explore the relationship between the various features and turbulence.

The key features influencing turbulence are as follows:

- Wind speed: Turbulence is worse with higher wind speeds
- Temperature: Turbulence is worse with higher temperatures
- Season: Turbulence is worse during winter and calmest during summer
- Time of day: Turbulence is worse during the day and calmest during the night/sunrise
- Ground elevation: Turbulence is worse when the ground is elevated
- Flight level: Turbulence is calmest at cruising altitude (30,000-40,000 ft)
- Weight class: Turbulence is worse for smaller aircrafts

Model

Four machine learning (ML) models were built and evaluated. The aim of these ML models is to learn from the dataset and classify the degree of turbulence into one of these categories: LIGHT, MODERATE, or SEVERE, by identifying patterns in the dataset.

The following table summarizes the performance of the models:

Model	Train Accuracy	Test Accuracy	Train-test Diff	Runtime
Decision Tree	99.66%	69.84%	29.82%	20.6s
Random Forest	81.67%	81.58%	0.09%	1m 50s
XGBoost	81.85%	81.58%	0.27%	1m 28s
CatBoost	81.66%	81.57%	0.09%	3m 54s

By evaluating the trade-offs between train and test accuracy, model discrepancies, and runtime, XGBoost has been identified as the best model for demonstrating superior performance with no significant overfitting.

Recommendations and Conclusion

This analysis is proof of concept that turbulence severity can be predicted relatively accurately using machine learning models. By utilizing a mega dataset of historical turbulence (over 1.5 million observations), the model uniquely predicts turbulence using ground-based meteorological data instead of data from plane sensors, and without traditional turbulence metrics commonly studied in the literature.

However, it is recommended that this analysis be repeated with real-time data from plane sensors to enhance its accuracy. Ground-based meteorological data may not be reflective of atmospheric conditions at various altitudes, and plane sensors also provide additional information on weather conditions in the air (e.g., wind shear, vertical wind gust, etc.). Ultimately, it is recommended that airlines integrate machine learning models into flight plans and pilot dashboards to help predict turbulence in real-time, in order to achieve smoother and safer flights for all.

References

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